# Multi-Channel Linguistic "Urgency" Detector

Full-Stack Product Requirements Document

SMS | WhatsApp | Email | AI/ML | NLP | REST API

Document Information

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|---|---|---|---|
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# 1. Project Overview & Vision

What we're building and why

## 1.1 Product Summary

The **Multi-Channel Linguistic Urgency Detector (MCLUD)** is an AI-powered scam-detection platform that analyses the *psychological pressure patterns* embedded in digital communications — SMS, WhatsApp, and Email — to identify and explain coercive messaging before it causes harm.

Unlike keyword-blacklist tools, MCLUD identifies the **tone of coercion**: artificial scarcity, authority impersonation, legal threat fabrication, and countdown pressure. Every flagged message receives a visual **Pressure Heatmap** that colour-codes sentences by manipulation tactic so users understand *why* a message is suspicious.

## 1.2 Goals & Success Metrics

| Goal | KPI | Target |
|---|---|---|
| Detect scam messages | Precision / Recall | ≥ 92% / ≥ 89% |
| Explain detections | Heatmap coverage | 100% of flagged msgs |
| Multi-channel support | Channels live at GA | SMS, WhatsApp, Email |
| Low-latency analysis | P95 API response | < 400 ms |
| Developer adoption | API integrations | 50 within 6 months |
| User trust / education | User rating (CSAT) | ≥ 4.2 / 5.0 |

## 1.3 User Personas

- **End Consumer** — Receives suspicious messages; wants quick, plain-English verdict + explanation.
- **Enterprise Security Team** — Needs bulk-scan API + analytics dashboard for corporate comms monitoring.
- **Platform Developer** — Integrates MCLUD REST API into messaging apps or SIEM tools.
- **Compliance Officer** — Requires audit logs, exportable reports, and GDPR/PDPA evidence trail.

# 2. System Architecture

High-level component topology

MCLUD follows a **microservices architecture** deployed on Kubernetes. The system separates ingestion, ML inference, API gateway, and frontend into independently scalable services connected via an internal message broker.

| Layer | Component | Responsibility | Tech |
|---|---|---|---|
| Ingestion | Channel Adapters | Normalise SMS/WhatsApp/Email to unified JSON | Python, Twilio SDK, IMAP |
| Processing | NLP Pre-processor | Tokenise, sentence-split, embed text | spaCy, Sentence-Transformers |
| ML | Classifier Service | Predict scam probability + tactic labels | PyTorch / scikit-learn |
| ML | Heatmap Generator | SHAP-based sentence attribution scores | SHAP, transformers |
| API | REST Gateway | Auth, rate-limit, route to services | FastAPI, Nginx |
| Storage | Primary DB | Messages, labels, audit logs | PostgreSQL 16 |
| Storage | Cache | Session tokens, hot inference results | Redis 7 |
| Storage | Object Store | Raw message archives, model artefacts | MinIO / S3 |
| Frontend | Web App | Dashboard, heatmap viewer, analytics | React 18, Tailwind CSS |
| Observability | Monitoring Stack | Metrics, traces, alerting | Prometheus, Grafana, Sentry |

## 2.1 Data Flow

1. Message arrives via Channel Adapter (webhook / IMAP poll / SDK callback).
2. Adapter normalises to **UnifiedMessage** schema and publishes to Kafka topic messages.raw.
3. NLP Pre-processor consumes, tokenises, sentence-splits, and produces embeddings.
4. Classifier Service scores the message and each sentence for tactic labels.
5. Heatmap Generator computes per-sentence SHAP attribution values.
6. Results written to PostgreSQL; enriched event published to messages.scored.
7. REST Gateway exposes result to frontend / third-party consumer.

# 3. Backend Requirements

Services, APIs, models, and data

## 3.1 Data Ingestion & Pre-processing

| ID | Requirement | Priority | Notes |
|---|---|---|---|
| BE-ING-01 | Accept webhooks from Twilio (SMS/WhatsApp) with HMAC-SHA256 signature verification. | HIGH | Replay attack window 300 s |
| BE-ING-02 | Poll IMAP/SMTP mailboxes every 60 s; support OAuth2 (Gmail, Outlook). | HIGH | |
| BE-ING-03 | Normalise all messages to UnifiedMessage JSON (id, channel, sender, body, timestamp, metadata). | HIGH | Schema v1.2 |
| BE-ING-04 | Strip HTML tags, decode MIME parts, handle base64 attachments. | HIGH | Use BeautifulSoup4 |
| BE-ING-05 | Detect language; pass only supported languages (EN, MS, ZH) to classifier; queue others. | MED | langdetect lib |
| BE-ING-06 | Deduplicate messages within 24 h window by content hash. | MED | Redis bloom filter |
| BE-ING-07 | Support bulk CSV/JSON upload endpoint (max 10 MB, 5 000 msgs per batch). | MED | Async job queue |

## 3.2 NLP & ML Classification Engine

### Model Architecture

A fine-tuned **DeBERTa-v3-base** transformer trained on the Fraudulent Email Corpus (Kaggle), SMS Spam Collection, and an internally labelled WhatsApp dataset. The model outputs: (a) binary scam/legit classification with confidence score, (b) multi-label tactic classification per sentence.

| Tactic Label | Description | Example Signal Phrases |
|---|---|---|
| ARTIFICIAL_SCARCITY | False urgency through limited supply/time | "Only 2 hours left", "Last chance" |
| AUTHORITY_IMPERSONATION | Fake government, bank, or platform identity | "Your bank account", "LHDN notice" |
| LEGAL_THREAT | Fabricated legal or criminal consequences | "Legal action", "warrant for arrest" |
| ACCOUNT_DEACTIVATION | Threat to disable account / access | "suspended", "verify immediately" |
| PRIZE_LURE | Unexpected reward requiring action | "You have won", "claim your prize" |
| PERSONAL_DATA_REQUEST | Soliciting credentials or personal details | "confirm your OTP", "enter IC" |
| ISOLATION_TACTIC | Instruction to keep secret / act alone | "tell no one", "private matter" |

### ML Requirements

| ID | Requirement | Priority | Metric |
|---|---|---|---|
| BE-ML-01 | Overall scam detection F1-score on held-out test set. | HIGH | F1 ≥ 0.91 |
| BE-ML-02 | Per-sentence tactic classification macro-F1. | HIGH | ≥ 0.85 |
| BE-ML-03 | SHAP attribution values computed for every sentence. | HIGH | 100 % coverage |

| ID | Requirement | Priority | Metric |
|---|---|---|---|
| BE-ML-04 | Model inference P95 latency (single message, CPU fallback). | HIGH | < 300 ms |
| BE-ML-05 | Model versioning: all artefacts stored with SHA-256 hash. | MED | MLflow tracking |
| BE-ML-06 | Automated retraining pipeline triggered weekly or on data drift. | MED | Evidently AI drift |
| BE-ML-07 | Calibrated probability outputs (Platt scaling). | MED | Brier score ≤ 0.08 |
| BE-ML-08 | Support for multi-lingual inference (EN, MS, ZH-simplified). | MED | Zero-shot + fine-tune |

## 3.3 API Layer

All endpoints are RESTful JSON over HTTPS. Authentication uses JWT (RS256) with 15-minute access tokens and 7-day refresh tokens. Rate limits are enforced per API key via Redis token-bucket algorithm.

| Method | Endpoint | Description | Auth |
|---|---|---|---|
| POST | /v1/analyse | Analyse a single message; returns scam score + heatmap. | API Key |
| POST | /v1/analyse/batch | Submit async batch job (CSV/JSON); returns job_id. | API Key |
| GET | /v1/jobs/{job_id} | Poll batch job status and retrieve results. | API Key |
| GET | /v1/messages/{id} | Retrieve stored analysis for a message. | JWT |
| GET | /v1/messages | List messages with filters (channel, date, score, tactic). | JWT |
| GET | /v1/analytics/summary | Aggregate stats: volumes, top tactics, trend lines. | JWT |
| POST | /v1/feedback | Submit human label correction for active learning. | JWT |
| GET | /v1/models/current | Current model version, training date, performance metrics. | JWT |
| POST | /auth/token | Exchange credentials for JWT access + refresh tokens. | Public |
| POST | /auth/refresh | Refresh access token. | Refresh JWT |
| GET | /health | Liveness + readiness probe. | Public |

### Sample Request/Response — POST /v1/analyse

```
Request body (JSON): { "channel": "sms", "body": "URGENT: Your Maybank account will be suspended in 2 hours. Click http://mybnk-verify.xyz to verify your identity.", "sender": "+60123456789" }
Response (200 OK): { "message_id": "msg_abc123", "scam_score": 0.97, "verdict": "SCAM", "tactics_detected": ["AUTHORITY_IMPERSONATION", "ARTIFICIAL_SCARCITY", "ACCOUNT_DEACTIVATION"], "heatmap": [ { "sentence": "URGENT: Your Maybank account will be suspended in 2 hours.", "tactics": ["AUTHORITY_IMPERSONATION","ACCOUNT_DEACTIVATION"], "intensity": 0.94 }, ... ], "processing_ms": 218 }
```

## 3.4 Database Schema (Key Tables)

| Table | Key Columns | Purpose |
|---|---|---|
| messages | id, channel, sender_hash, body_encrypted, received_at, scam_score, verdict | Primary message store (PII encrypted at rest) |
| heatmap_results | id, message_id, sentence_idx, sentence_text, tactics[], intensity, shap_values | Per-sentence analysis output |
| tactic_labels | id, code, display_name, description, example_phrases[] | Reference: tactic taxonomy |
| feedback | id, message_id, user_id, correct_verdict, correct_tactics[], created_at | Human corrections for active learning |

| Table | Key Columns | Purpose |
|---|---|---|
| api_keys | id, owner_id, key_hash, scopes[], rate_limit_rpm, created_at, revoked_at | API key management |
| audit_logs | id, actor_id, action, resource_type, resource_id, ip_hash, ts | GDPR/PDPA compliance trail |
| model_versions | id, artefact_path, sha256, f1_score, trained_at, deployed_at, active | ML model registry |

# 4. Frontend Requirements

React SPA — Dashboard, Heatmap, Analytics

## 4.1 Dashboard & Message Upload Interface

| ID | Requirement | Priority |
|---|---|---|
| FE-DASH-01 | Landing screen: single-message analyser — paste or type text, select channel, click Analyse. | HIGH |
| FE-DASH-02 | Display verdict badge (SCAM / SUSPICIOUS / SAFE) with colour coding (red/amber/green) and confidence percentage. | HIGH |
| FE-DASH-03 | Bulk upload panel: drag-and-drop CSV/JSON; show live progress bar with job polling. | HIGH |
| FE-DASH-04 | Message history table with sortable columns: date, channel, score, top tactic, actions. | HIGH |
| FE-DASH-05 | Filters: channel, date range, verdict, tactic type; persist filter state in URL params. | MED |
| FE-DASH-06 | Responsive design — full functionality on 320px–2560px viewports. | HIGH |
| FE-DASH-07 | Dark mode / light mode toggle; persist preference in localStorage. | MED |
| FE-DASH-08 | Onboarding tour (Shepherd.js or Intro.js) for first-time users. | LOW |

## 4.2 Pressure Heatmap Viewer

The heatmap is the core differentiating UI. Each sentence in the analysed message is rendered inline with background intensity proportional to its SHAP attribution score. Hovering or tapping a sentence reveals a tooltip with the assigned tactic labels and an educational explanation.

| ID | Requirement | Priority |
|---|---|---|
| FE-HM-01 | Render message body sentence-by-sentence with background colour intensity mapped to SHAP score (white → amber – | HIGH |
| FE-HM-02 | Each sentence displays a tactic chip (e.g. 'AUTHORITY IMPERSONATION') on hover/tap. | HIGH |
| FE-HM-03 | Tooltip content: tactic name, plain-English explanation (≤ 40 words), SHAP intensity value. | HIGH |
| FE-HM-04 | Tactic Legend panel listing all detected tactics with colour swatches and counts | HIGH |
| FE-HM-05 | Export heatmap as PNG image or PDF report (single-page) via browser print or canvas snapshot. | MED |
| FE-HM-06 | Share-link generation: encode message_id in URL; shareable for 72 h (token-gated). | MED |
| FE-HM-07 | Feedback widget: thumbs-up/down per sentence label; submitted to /v1/feedback | MED |
| FE-HM-08 | Accessibility: ARIA labels on all sentence spans; keyboard navigation between sentences. | HIGH |
| FE-HM-09 | Animated entry — sentences fade in sequentially (150 ms stagger) to guide reading | LOW |

## 4.3 Analytics & Reporting

| ID | Requirement | Priority |
|---|---|---|
| FE-ANL-01 | Line chart: daily/weekly scam volume trend by channel (Recharts). | HIGH |
| FE-ANL-02 | Bar chart: tactic frequency breakdown across all messages. | HIGH |
| FE-ANL-03 | Pie / donut chart: channel distribution (SMS vs WhatsApp vs Email). | MED |
| FE-ANL-04 | Heatmap calendar: message volume per day (GitHub contribution style). | MED |
| FE-ANL-05 | Date-range picker for all charts; support: today, 7d, 30d, 90d, custom. | HIGH |
| FE-ANL-06 | Export analytics data to CSV; export charts to PNG. | MED |
| FE-ANL-07 | Real-time dashboard refresh every 60 s via Server-Sent Events. | MED |
| FE-ANL-08 | Anomaly alert banner when scam rate spikes > 2× 7-day average. | HIGH |

## 4.4 Account & Settings

| ID | Requirement | Priority |
|---|---|---|
| FE-SET-01 | Registration / login (email + password; Google OAuth optional). | HIGH |
| FE-SET-02 | API key management: create, name, copy, revoke keys; show usage stats. | HIGH |
| FE-SET-03 | Channel configuration: connect Twilio account, IMAP mailbox, WhatsApp Business. | HIGH |
| FE-SET-04 | Notification preferences: email / webhook alerts on SCAM verdict. | MED |
| FE-SET-05 | Data retention settings: auto-delete messages after N days. | MED |
| FE-SET-06 | Usage billing dashboard (if SaaS tier): API calls consumed vs plan limit. | LOW |

# 5. Functional Requirements (Use Cases)

User stories and acceptance criteria

### UC-01 — Analyse Single Message

User pastes a suspicious SMS into the text box and selects channel. System returns verdict, overall score, and a heatmap with per-sentence tactic labels within 5 seconds.

**Acceptance Criteria:**

✓ Message body is not empty
✓ Response arrives in < 5 s
✓ Heatmap contains ≥ 1 sentence entry
✓ Verdict is one of SCAM / SUSPICIOUS / SAFE

### UC-02 — Bulk Batch Scan

Enterprise user uploads a CSV with 1 000 messages. System processes asynchronously, user polls for completion and downloads a results CSV.

**Acceptance Criteria:**

✓ File ≤ 10 MB and ≤ 5 000 rows
✓ Job status cycles: QUEUED → PROCESSING → DONE
✓ Results CSV includes all input columns + score/verdict/tactics columns
✓ Failed rows listed in errors[] array with reason

### UC-03 — Channel Auto-Scan

User connects their IMAP email. System polls inbox every 60 s and auto-analyses new messages, flagging SCAM verdicts in the dashboard.

**Acceptance Criteria:**

✓ OAuth2 flow completes without storing cleartext credentials
✓ New emails detected within 90 s of arrival
✓ SCAM verdict triggers in-app + email notification (if enabled)

### UC-04 — Submit Feedback

User disagrees with a tactic label on a sentence and marks it incorrect via the thumbs-down widget.

**Acceptance Criteria:**

✓ Feedback stored with message_id, sentence_idx, user_id
✓ Confirmation toast shown within 500 ms
✓ Feedback excluded from public model training until review

# 6. Non-Functional Requirements

Performance, scalability, security, usability

| Category | Requirement | Target / Constraint |
|---|---|---|
| Performance | API P95 latency — single message analysis | < 400 ms |
| Performance | API P99 latency — single message analysis | < 1 200 ms |
| Performance | Batch job throughput | ≥ 500 messages / minute |
| Performance | Frontend Time-to-Interactive (TTI) | < 2.5 s on 4G |
| Scalability | Horizontal pod autoscaling on inference service | 2–20 replicas, CPU trigger 70% |
| Scalability | Kafka partition strategy for message throughput | ≥ 10 000 msg/s peak |
| Availability | API service uptime SLA | 99.9% monthly |
| Availability | Planned maintenance window | Sundays 02:00–04:00 UTC |
| Reliability | Zero message loss for ingestion pipeline | Kafka at-least-once + idempotent consumer |
| Security | All data in transit | TLS 1.3 minimum |
| Security | PII fields in database (sender, body) | AES-256 encrypted at rest |
| Security | OWASP Top 10 compliance | No Critical/High vulnerabilities at release |
| Privacy | GDPR / PDPA — right to erasure | Deletion within 30 days of request |
| Privacy | Message data retention default | 90 days (user-configurable) |
| Usability | WCAG 2.1 AA accessibility compliance | All interactive components |
| Usability | First-time user to first analysis | < 3 minutes (no training required) |
| Maintainability | Code test coverage (unit + integration) | ≥ 80% |
| Maintainability | API documentation | OpenAPI 3.1 spec auto-generated |
| Internationalisation | UI language support at launch | English, Malay, Mandarin (Simplified) |

# 7. Technology Stack

Approved libraries, frameworks, and infrastructure

| Layer | Technology | Version | Justification |
| --- | --- | --- | --- |
| ML / NLP | PyTorch | 2.3 | DeBERTa fine-tuning |
| ML / NLP | Hugging Face Transformers | 4.41 | Model hub + tokenisers |
| ML / NLP | spaCy | 3.7 | Sentence segmentation |
| ML / NLP | SHAP | 0.45 | Explainability values |
| ML / NLP | scikit-learn | 1.5 | Calibration, metrics |
| ML / NLP | MLflow | 2.13 | Experiment tracking |
| Backend | Python | 3.12 | Primary language |
| Backend | FastAPI | 0.111 | Async REST framework |
| Backend | Pydantic v2 | 2.7 | Schema validation |
| Backend | Celery + Redis | 5.4 / 7 | Task queue |
| Backend | Apache Kafka | 3.7 | Event streaming |
| Backend | SQLAlchemy + Alembic | 2.0 | ORM + migrations |
| Backend | PostgreSQL | 16 | Primary RDBMS |
| Backend | MinIO | Latest | S3-compatible object store |
| Frontend | React | 18 | UI framework |
| Frontend | TypeScript | 5.4 | Type safety |
| Frontend | Tailwind CSS | 3.4 | Utility-first styling |
| Frontend | Recharts | 2.12 | Chart components |
| Frontend | Tanstack Query | 5 | API state management |
| Frontend | Zustand | 4 | Global state |
| Frontend | Vite | 5 | Build tool |
| Infrastructure | Kubernetes (K8s) | 1.30 | Container orchestration |
| Infrastructure | Docker | 26 | Containerisation |
| Infrastructure | Nginx | 1.26 | Reverse proxy / ingress |
| Infrastructure | Prometheus + Grafana | Latest | Metrics & dashboards |
| Infrastructure | Sentry | Latest | Error tracking |
| Infrastructure | GitHub Actions | Latest | CI/CD pipelines |

# 8. Data Requirements

Training datasets, labelling, and pipelines

| Dataset | Source | Size (approx.) | Usage |
|---|---|---|---|
| Fraudulent Email Corpus | Kaggle | ~82 000 emails | Primary email training data |
| SMS Spam Collection | UCI / Kaggle | ~5 600 SMS | SMS baseline training |
| Nigerian Prince Email Set | Apache SpamAssassin | ~6 000 emails | Authority impersonation patterns |
| Internal WhatsApp Samples | Crowd-sourced, manual labels | ~3 000 msgs | WhatsApp-specific patterns |
| TREC 2007 Spam Track | NIST TREC | ~75 000 emails | Negative (ham) examples |
| Augmented Synthetic Malay | GPT-4 + human review | ~5 000 msgs | Malay-language scam coverage |

**Labelling Protocol:** Dual annotator agreement required (Cohen's $\kappa \geq 0.75$). Disputed labels resolved by a third senior annotator. Labels exported in JSONL with sentence-level spans and tactic codes.

## 9. Security & Compliance

| Control | Implementation | Standard |
|---|---|---|
| Authentication | JWT RS256; rotate signing keys every 90 days | OWASP ASVS L2 |
| Authorisation | RBAC: roles Admin / Analyst / API-User | Least privilege |
| Input Sanitisation | Strip HTML, validate JSON schema at gateway | OWASP Top 10 A03 |
| Injection Prevention | Parameterised queries via SQLAlchemy ORM | CWE-89 |
| Secrets Management | HashiCorp Vault; no secrets in code or env files | NIST SP 800-57 |
| Audit Logging | Immutable audit_logs table; 12-month retention | PDPA 2010 (MY) / GDPR |
| Data Masking | Sender phone/email hashed (SHA-256 + salt) before storage | GDPR Art. 25 |
| Vulnerability Scanning | Trivy on Docker images in CI; Dependabot alerts | NIST NVD |
| Penetration Testing | Annual third-party pentest + quarterly DAST scans | ISO 27001 A.12.6 |
| Breach Notification | Notify affected users within 72 h of confirmed breach | GDPR Art. 33 |

# 10. Testing Strategy

Unit, integration, ML evaluation, and E2E

| Test Type | Scope | Tool | Pass Criteria |
|---|---|---|---|
| Unit | Individual functions, parsers, validators | pytest | 100% of critical paths; ≥ 80% overall coverage |
| Integration | API endpoints ↔ DB ↔ Redis | pytest + httpx | All endpoints 2xx/4xx contract verified |
| ML Evaluation | Classifier F1, calibration, SHAP correctness | pytest + custom harness | F1 ≥ 0.91; Brier ≤ 0.08; SHAP sum = model output |
| Contract | API schema vs OpenAPI spec | Schemathesis | Zero schema violations |
| Load | 200 concurrent analyse requests for 5 min | k6 | P95 < 400 ms; error rate < 0.1% |
| Security | SAST + DAST scan | Bandit + OWASP ZAP | No Critical/High open findings |
| Accessibility | Screen-reader, keyboard navigation | axe-core + manual | WCAG 2.1 AA — 0 violations |
| E2E | Full user journeys: single-scan, bulk, settings | Playwright | All happy-path journeys pass in CI |
| Regression | ML model performance on frozen benchmark set | pytest + mlflow | No degradation vs previous deployed version |

# 11. Deployment & DevOps

CI/CD, environments, and infrastructure-as-code

## 11.1 Environments

| Environment | Purpose | Deployment Trigger | Data |
|---|---|---|---|
| Development | Local dev & rapid iteration | Manual / hot-reload | Anonymised fixtures |
| Staging | Integration testing & QA sign-off | Push to main branch | Anonymised prod snapshot |
| Production | Live service | Tagged release (semver) | Live encrypted data |
| DR (passive) | Disaster recovery hot standby | Replication from prod | Replicated prod data |

## 11.2 CI/CD Pipeline Stages

**Stage 01.** Lint & Format — Ruff (Python), ESLint + Prettier (TypeScript).
**Stage 02.** Unit & Integration Tests — pytest + Playwright in parallel matrix.
**Stage 03.** ML Regression — compare model metrics vs baseline on benchmark set.
**Stage 04.** Docker Build — multi-stage image build; Trivy vulnerability scan.
**Stage 05.** Staging Deploy — Helm upgrade to staging namespace; smoke tests.
**Stage 06.** Load Test — k6 100-VU 3-minute ramp against staging.
**Stage 07.** Manual Approval Gate — product owner sign-off for production.
**Stage 08.** Production Deploy — blue/green rollout with 10% canary for 15 min.
**Stage 09.** Post-deploy Health Check — readiness probe + synthetic transaction.

# 12. Milestones & Timeline (MVP → GA)

6-month delivery roadmap

| Phase | Duration | Key Deliverables | Exit Criteria |
|---|---|---|---|
| Phase 0 — Foundation | Week 1–2 | Repo structure, CI skeleton, DB schema v1, ADRs written | All PR templates & branch policies in place |
| Phase 1 — ML Baseline | Week 3–6 | DeBERTa fine-tuned on email corpus; F1 ≥ 0.88 on test set; SHAP pipeline | Model artefact registered in MLflow; unit tests green |
| Phase 2 — Backend MVP | Week 5–8 | FastAPI service: /analyse endpoint live; Twilio SMS webhook; PostgreSQL schema | Postman collection all 200 OK; Kafka pipeline end-to-en |
| Phase 3 — Frontend MVP | Week 7–10 | React dashboard: single-scan UI + heatmap viewer; auth flows | E2E Playwright suite passes; WCAG audit clean |
| Phase 4 — Multi-Channel | Week 9–12 | WhatsApp Business, IMAP email adapters; language detection | All three channels verified in staging |
| Phase 5 — Analytics | Week 11–14 | Analytics charts; bulk batch API; admin settings panel | Load test passes (k6); bulk 1 000 msgs < 3 min |
| Phase 6 — Hardening | Week 13–16 | Security pentest remediation; GDPR controls; DR drill; docs | 0 Critical/High vulns; DR RTO verified < 1 h |
| Phase 7 — Beta | Week 17–20 | Private beta with 5 enterprise customers; feedback integration | NPS ≥ 30; ≤ 5 P1 bugs open |
| Phase 8 — GA Release | Week 21–24 | Public launch; SLA active; on-call rotation; billing (if SaaS) | 99.9% uptime SLA signed; support SLA documented |

# 13. Risks & Mitigations

Identified risks and response strategies

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| ML model under-performs on Malay/ZH text due to limited training data | HIGH | HIGH | Augment with GPT-4 synthetic data; deploy multilingual base model (mDeBERTa) |
| Twilio or WhatsApp API terms change, breaking channel adapters | MED | HIGH | Abstract channel adapters behind interface; monitor vendor changelogs |
| High-volume batch jobs overwhelm inference service | MED | HIGH | Auto-scale Kubernetes pods; implement back-pressure via Kafka consumer groups |
| Adversarial scammers adapt phrasing to evade classifier | HIGH | MED | Weekly retraining pipeline; monitor false-negative rate in production |
| GDPR/PDPA breach due to insufficient PII handling | LOW | HIGH | Encrypt at rest; hash sender identifiers; legal review before launch |
| Key engineer attrition causing knowledge loss | MED | MED | ADRs for all major decisions; pair-programming culture; runbook for all services |
| Third-party ML library (Hugging Face) introduces breaking changes | LOW | MED | Pin dependency versions; isolated Docker images; update in dedicated branch |

## 14. Glossary

**SHAP**: SHapley Additive exPlanations — a game-theory method that assigns each input feature a contribution score to the model's output.

**DeBERTa**: Decoding-enhanced BERT with Disentangled Attention — a transformer model by Microsoft that outperforms BERT/RoBERTa on NLU tasks.

**Pressure Heatmap**: A visual overlay on a message body that colour-codes each sentence by manipulation intensity and labels the tactic used.

**Tactic**: A named psychological manipulation pattern (e.g. ARTIFICIAL_SCARCITY) used to coerce the recipient into taking an action.

**UnifiedMessage**: The internal JSON schema that normalises messages from all channels (SMS, WhatsApp, Email) into a single structure.

**PDPA**: Personal Data Protection Act 2010 (Malaysia) — governs the collection, storage, and use of personal data.

**MLflow**: An open-source platform for managing the ML lifecycle, including experiment tracking and model registry.

**Bloom Filter**: A probabilistic data structure used for fast, memory-efficient membership testing (here: deduplication check).

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Document prepared April 2026 · Multi-Channel Linguistic Urgency Detector v1.0.0 · All requirements subject to change pending stakeholder review.